

**UNIVERSITY OF MADRAS**  
**M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE**  
**SYLLABUS WITH EFFECT FROM 2023-2024**

|                        |                                     |          |        |
|------------------------|-------------------------------------|----------|--------|
| Title of the Paper     | <b>Distributed Database Systems</b> |          |        |
| Elective – II - Theory | I Year & I Semester                 | Credit:3 | 436E1F |

**Objectives:**

To introduction students to Distributed DBMS and associated problems.

To make students understand various algorithms and techniques for managing distributed database.

To understand theoretical and practical aspects of distributed database systems.

To study and identify various issues related to the development of distributed database system.

To make students understand Transaction Management & Compare various approaches to concurrency control in Distributed database

**Outcomes:**

|                                                                      |                                                                        |        |
|----------------------------------------------------------------------|------------------------------------------------------------------------|--------|
| 1.                                                                   | Apply various fragmentation techniques given a problem                 | K1     |
| 2.                                                                   | Analyse and calculate the cost of enforcing semantic integrity control | K2, K3 |
| 3.                                                                   | Use the steps of query processing                                      | K4     |
| 4.                                                                   | Apply optimization techniques are applies to Distributed Database      | K4, K5 |
| 5.                                                                   | Apply effectively Query Optimization Algorithms                        | K5, K6 |
| K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create |                                                                        |        |

Unit-I: Introduction: Distributed Data Processing, Distributed Database Systems, Promises of DDBSs, Complicating factors, Problem areas Overview of RDBMS: Concepts, Integrity, Normalization - Distributed DBMS Architecture: Autonomy, Distribution, Heterogeneity DDBMS Architecture – Client/Server, Peer to peer, MDBS

Unit-II: Data Distribution Alternatives: Design Alternatives – localized data, distributed data Fragmentation – Vertical, Horizontal (primary & derived), hybrid, general guidelines, correctness rules Distribution transparency – location, fragmentation, replication Impact of distribution on user queries – No Global Data Dictionary (GDD), GDD containing location information Example on fragmentation

Unit-III: Semantic Data Control: View Management, Authentication – database authentication, OS authentication, Access Rights, Semantic Integrity Control – Centralized & Distributed, Cost of enforcing semantic integrity - : Query Processing: Query Processing Problem, Layers of Query Processing Query Processing in Centralized Systems – Parsing & Translation, Optimization, Code generation, Example Query Processing in Distributed Systems – Mapping global query to local, Optimization,

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Unit-IV: Optimization of Distributed Queries: Query Optimization, Centralized Query Optimization, Join Ordering Distributed Query Optimization Algorithms - Distributed Transaction Management & Concurrency Control: Transaction concept, ACID property, Objectives of transaction management, Types of transactions, Objectives of Distributed Concurrency Control, Concurrency Control anomalies, Methods of concurrency control, Serializability and recoverability, Distributed Serializability, Enhanced lock based and timestamp based protocols, Multiple granularity, Multi version schemes, Optimistic Concurrency Control techniques

Unit-V: Distributed Deadlock & Recovery: Deadlock concept, Deadlock in Centralized systems, Deadlock in Distributed Systems – Detection, Prevention, Avoidance, Wait-Die Algorithm, Wound-Wait algorithm Recovery in DBMS - Types of Failure, Methods to control failure, Different techniques of recoverability, Write- Ahead logging Protocol, Advanced recovery techniques- Shadow Paging, Fuzzy checkpoint, ARIES, RAID levels, Two Phase and Three Phase commit protocols

**Recommended Texts:**

1. Ozsu; Principles of Distributed Database Systems; Springer; 4th edition;2020

**Reference Books:**

1. Rahimi &Haug; Distributed Database Management Systems;Wiley;2010
2. Distributed Database Systems, Chanda Ray, Pearson Publication
3. Sachin Deshpande; Distributed Databases; Dreamtech;2014

**Web References:**

1. <https://www.youtube.com/watch?v=dIBVWMdGhqw&list=PLUJ7JmcrTifBROWODSG8wgyl20XgBuE-N>

Mapping with Programme Outcomes:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| CO1 | M   | S   | M   | S   | S   | M   | L   | M   | L   | S    |
| CO2 | S   | M   | S   | L   | M   | L   | M   | L   | M   | S    |
| CO3 | S   | S   | L   | M   | S   | S   | L   | M   | L   | M    |
| CO4 | M   | L   | S   | M   | M   | L   | S   | L   | M   | S    |
| CO5 | S   | S   | M   | S   | L   | M   | L   | M   | S   | L    |

S-Strong M-Medium L-Low